Project Initialization and Planning Phase

Date	7 July 2026
Team ID	XXXX
Project Title	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

---

1. Project Proposal Description Block

text

The proposed solution aims to transform the loan eligibility assessment process using Machine Learning to improve efficiency, consistency, and decision-making. Traditional credit and loan application screening often involves manual verification, making the process time-consuming and prone to human error. The proposed system automates the preliminary eligibility assessment by analyzing applicant demographic, financial, and property-related information using machine learning algorithms. The application provides instant eligibility predictions, confidence scores, and personalized risk recommendations through a user-friendly web interface, helping both applicants and financial institutions make informed decisions.

---

2. Project Proposal Scope Table

Section	Details
Objective	The primary objective is to develop an intelligent web-based system that predicts loan approval eligibility using machine learning techniques, enabling faster, more accurate, and consistent preliminary assessments.
Scope	The project covers data preprocessing, SMOTE balancing, feature engineering, machine learning model training, model evaluation, and deployment of a Flask-based web application that provides real-time loan eligibility predictions.

3. Problem Statement Table

Section	Details
Description	Banks and financial institutions receive a large number of loan applications every day. Manual screening is time-consuming, inconsistent, and may result in delays or incorrect decisions. Applicants also lack a reliable way to estimate their eligibility before submitting a formal application.
Impact	Automating the eligibility assessment process reduces manual effort, improves consistency in decision-making, speeds up application screening, enhances customer experience, and helps financial institutions identify high-risk applicants early.

4. Proposed Solution Table

Section	Details
Approach	Build a machine learning-based web application that analyzes applicant information, preprocesses the data, and predicts loan eligibility using a trained XGBoost classification model. The system provides real-time predictions through a Flask web interface.
Key Features	• Machine learning-based loan eligibility prediction. • Automated data preprocessing and feature scaling. • Comparison of multiple classification algorithms (Logistic Regression, Decision Tree, Random Forest, KNN, SVM, and XGBoost). • Real-time eligibility prediction with confidence score and risk level. • Personalized recommendations for applicants (Credit Warning & DTI Caps). • Responsive Flask web application with dynamic range sliders. • Cloud deployment using Render.

5. Resource Requirements Table

Resource Type	Description	Specification / Allocation
Hardware	Computing Resources	Intel Core i3/i5/i7 Processor or equivalent
	Memory	Minimum 8 GB RAM

Resource Type	Description	Specification / Allocation
	Storage	Minimum 20 GB Free Disk Space
Software	Framework	Flask
	Programming Language	Python 3.10+
	Libraries	Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Joblib/Pickle
	Development Environment	Visual Studio Code / PyCharm
	Version Control	Git & GitHub
	Deployment Platform	Render with Gunicorn
Data	Dataset	loan_data.csv
	Source	Kaggle (Loan Eligibility Prediction Dataset)
	Format	CSV
	Records	614 applicant records
	Features	Selected demographic, financial, and property attributes (11 features) used for machine learning prediction.

10:26 PM